Constellation

10 CFR 50.73

LG-24-067

June 18, 2024

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Limerick Generating Station, Unit 1
Renewed Facility Operating License No. NPF-39
NRC Docket No. 50-352

Subject: LER 2024-003-00 Core Operating Limits Report Thermal Limits Exceeded

In accordance with the requirements of 10 CFR 50.73(a)(2)(i)(B), Limerick Generating Station hereby submits the enclosed Licensee Event Report.

There are no commitments contained in this letter.

If you have any questions, please contact Jordan Rajan at (610) 718-3400.

Respectfully,

Michael F. Gillin

Michael F. Gillin
Vice President – Limerick Generating Station
Constellation Energy Generation, LLC

cc: Administrator Region I, USNRC
USNRC Senior Resident Inspector, Limerick Generating Station

NRC FORM 366 (04-02-2024)	U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104	EXPIRES: 04/30/2027

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		2. Docket Number	3. Page
Limerick Generating Station Unit 1	☒ 050 ☐ 052	352	1 OF 3

4. Title
Core Operating Limits Report Thermal Limit Exceeded for Unit 1 Fuel Bundle GGK123

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
04	22	24	2024	- 003 -	00	06	18	2024	Facility Name	☐ 052	Docket Number

9. Operating Mode	10. Power Level
4	0

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☑ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Jordan Rajan	610-718-3400

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	AC	ROD	G080	Yes					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☑ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On April 12, 2024, while performing a water rod inspection of fuel bundle GGK123 during a planned refuel outage, it was identified that several fuel spacers had propagated up within the bundle toward the upper tie plate. It was determined through analysis that spacer positioning had a direct effect on the Minimum Critical Power Ratio (MCPR). On April 22, 2024, an analysis of the operating conditions by Global Nuclear Fuel (GNF) of fuel bundle GGK123 was reviewed and accepted by the station. The report confirmed that, during the previous cycle, the Operating Limit MCPR exceeded limits set by Technical Specifications (TS) 3.2.3. Since the Operating Limit MCPR was exceeded for longer than allowed by TS Limiting Condition for Operation (LCO) 3.2.3, this event is reportable as a condition prohibited by TS per 10 CFR 50.73(a)(2)(i)(B).

The fuel bundle spacers shifted out of position over the previous operating cycle. This condition was identified during the inspections in response to GNF 10 CFR Part 21 communication SC 22-04, GNF3 Raised Water Rod. This fuel bundle has been removed from the reactor core, is currently located in the spent fuel pool, and will not be re-loaded into the core. There was no impact to the health and safety of the public or plant personnel because the Safety Limit MCPR was not exceeded in the previous cycle operation.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
			YEAR	SEQUENTIAL NUMBER	REV NO.
Limerick Generating Station Unit 1	☒ 050 / ☐ 052	352	2024	- 003	- 00

NARRATIVE

Condition Prior to Event

On April 22, 2024, prior to the determination that fuel bundle GGK123 had exceeded a thermal limit, Limerick Generating Station Unit 1 was in Operational Condition (OPCON) 4 (Cold Shutdown) with reactor coolant temperature of 132 degrees Fahrenheit.

Event Description

On April 12, 2024, during planned in-core fuel inspections for mis-oriented water rods of susceptible GNF3 fuel bundles [EIIS System AC] in response to GNF 10 CFR Part 21 communication SC 22-04, an unexpected condition for fuel bundle GGK123 was discovered where four spacers visibly collocated below the upper tie plate. The four spacers located in the bottom portion of the fuel bundle were found at their expected (installed) axial locations. Fuel bundle GGK123 was permanently discharged to the spent fuel pool.

Fuel vendor GNF evaluated the impact of this condition for the previous operating cycle. The evaluation conservatively assumed the spacers were displaced for all but the first week of the cycle. The inspection results showed the spacer contact marks indicated prolonged contact with the fuel indicating the spacers did not move early in the cycle. On April 22, 2024, the station completed the evaluation for fuel bundle GGK123.

Analysis of the Event

The evaluation report determined that:

- There were no penalties or adjustments associated with Linear Heat Generation Rate (LHGR) and Average Planar Linear Heat Generation Rate (APLHGR) and no discernible impact to thermal limits and margins.
- There was no discernible impact to thermal limits and margins for co-resident fuel in the Unit 1 reactor core during the previous cycle of operation.
- The Operating Limit Minimum Critical Power Ratio (MCPR) exceeded limits set by TS 3.2.3 for the previous cycle.
- The Safety Limit MCPR requirements set by TS 2.1.2 were never violated for the previous cycle.

The report determined that the requirements of TS LCO 3.2.3 MCPR were not met for Unit 1 Cycle 20. TS LCO 3.2.3 requires all MCPR values to be greater than or equal to the MCPR operating limits specified in the Core Operating Limits Report (COLR). If the MCPR value is less than the MCPR value in the COLR, TS LCO 3.2.3 Action b requires corrective actions to be initiated within 15 minutes and the MCPR value restored to within the required limits within two-hours or reduce thermal power to less than 25 percent within the next 4 hours. Since the station conservatively assumed the condition existed for most of the operating cycle, the station exceeded the two-hour requirement to restore MCPR within limits and did not reduce thermal power to less than 25 percent within the next 4 hours. Therefore, this event is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by the plant's TS.

Safety Consequences

There were no actual safety consequences to this condition. The MCPR Safety Limit was not exceeded and there was no breach of fuel cladding for fuel bundle GGK123.

NARRATIVE

Cause of the Condition

The cause of the spacer relocation was determined to be the mis-oriented water rod in the fuel bundle GGK123. Mis-orientation of the water rod would have occurred during fuel bundle fabrication. The mis-orientation caused the spacers to not be locked into position, causing them to propagate up the bundle due to hydraulic force during normal operation.

Corrective Actions

Corrective actions taken in response to the conditions were:

- Verified no plant off-gas signatures indicative of leaking fuel rods which verified fuel rod integrity had not been compromised.
- Performed full length peripheral visual inspection of de-channeled fuel bundle GGK123.
- Evaluated the impact to APLHGR, LHGR, and MCPR limits.
- Permanently discharged fuel bundle GGK123 to the spent fuel pool.
- Completed inspection on the remaining 267 fuel bundles in the Unit 1 susceptible population and confirmed no additional fuel bundles had a mis-oriented water rod.
- Confirmed no impact to Unit 2 due to no fuel bundles in the susceptible population.
- GNF implemented the following additional checks during the fuel fabrication process:
    - An in-process laser check on the orientation of the water rod tabs.
    - Removed the Quality at the Source process and reintroduced Qualify Inspectors to the bundle assembly tables.
    - Added a check for tab orientation at final inspection.
- Constellation Energy Generation, LLC (and Limerick Generating Station) enhanced the new fuel receipt inspection procedures to confirm proper water rod orientation on all new GNF3 fuel bundles.

Previous Occurrences

There were no previous occurrences.

Component Failure Data

System: Reactor Core Systems (AC)
Component: Unit 1 Fuel Bundle (ROD)
Manufacturer: Global Nuclear Fuel (G080)